

Question	Answer	Marks
9(a)	(two small) nuclei join together	M1
	to form one larger nucleus	A1
9(b)	line with a peak at $A \approx 56$	B1
	line with steep initial positive gradient on the left of peak and shallower negative gradient at all points to the right of peak and line does not return to 0 binding energy	B1
9(c)(i)	X shown at value of A to the right of the peak	B1
9(c)(ii)	Y shown at value of A close to 1	B1
9(d)	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$ $(= 2.94 \times 10^{-11} \text{ J})$	C1
	binding energy of Z = $[(1.25 + 1.81) \times 10^{-10}] - 2.94 \times 10^{-11}$ $(= 2.77 \times 10^{-10} \text{ J})$	C1
	nucleon number of Z = $93 + 139 + 2 - 1$ $(= 233)$	C1
	binding energy per nucleon = $(2.77 \times 10^{-10}) / (233 \times 1.60 \times 10^{-13})$ $= 7.43 \text{ MeV}$	A1

9702/43

Cambridge International AS & A Level – Mark Scheme

October/November 2023

PUBLISHED

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10(c)	ultrasound production; vibrational events; crystal	D1